SEA × OPENAI CODEX HACKATHON VIETNAM 2026

Guardline



An AI assistant that stops SPayLater fraud. It finds groups of fake accounts working together, takes safe actions on its own, and gets smarter from every decision your team makes.

THE PROBLEM

Fake orders that turn loans into cash

Fraudsters use many accounts to place fake orders at a shop they work with, take the cash, and never repay. Each order looks normal on its own.

BUILD DIRECTION

Autonomous & Adaptive AI

It handles small, undoable actions by itself and sends big decisions to people. When it finds a new trick, it writes a new rule, so next time it is caught fast.

THE AI WE APPLY

Built with OpenAI models

- An AI agent looks up orders, phones and addresses to check each case
- It gives a clear decision and shows the proof
- A fast model for easy cases, a smarter one for hard cases
- Simple checks run first, so most orders never need AI

Thao · Technical Product Manager **Thành** · Technical Lead **Hiếu** · AI Engineer

Monee wants fast growth with bad loans held at ~1%

US$11.1B
Loan book in Q2 2026, up 62% year on year

5.3M
First-time borrowers added in one quarter

>20%
of SPayLater loans used outside Shopee via QR

~1%
Bad loans (90+ days overdue): the line to hold

Fraud is the biggest threat to that goal

Cash-out: fake orders at a partner shop turn credit limits into cash that is never repaid. Shopee, Home Credit and MoMo have all warned users.

- **Hard to spot:** each order looks normal; fraud shows only across accounts, shops, phones and addresses
- **Keeps changing:** once a rule stops one trick, fraudsters switch
- **Hurts real customers:** blunt rules block families and nhà trọ renters, slowing growth
- **Slow and inconsistent:** cases are checked by hand

What we improve

Stop fraud before money is lost
Fraud caught before payout

Block fewer honest customers
Genuine users blocked

Cut investigation time
Minutes per case, decision consistency

Sources: Sea Q2 2026 earnings; Shopee, Home Credit and MoMo public warnings

One loop: detect, investigate, act safely, adapt

STEP 1

Detect

Rules, a link graph of buyers, shops, devices and addresses, and an anomaly score for patterns no rule covers.

STEP 2

Investigate

An AI agent works the whole cluster and writes a Vietnamese case report. All evidence cites a real record.

STEP 3

Act safely

Acts alone only on reversible steps. Credit calls go to the partner bank; large or novel cases go to a person.

STEP 4

Adapt

A confirmed new tactic becomes a drafted, backtested rule. A person promotes it; the next wave needs no AI.

STEP 5

Daily review

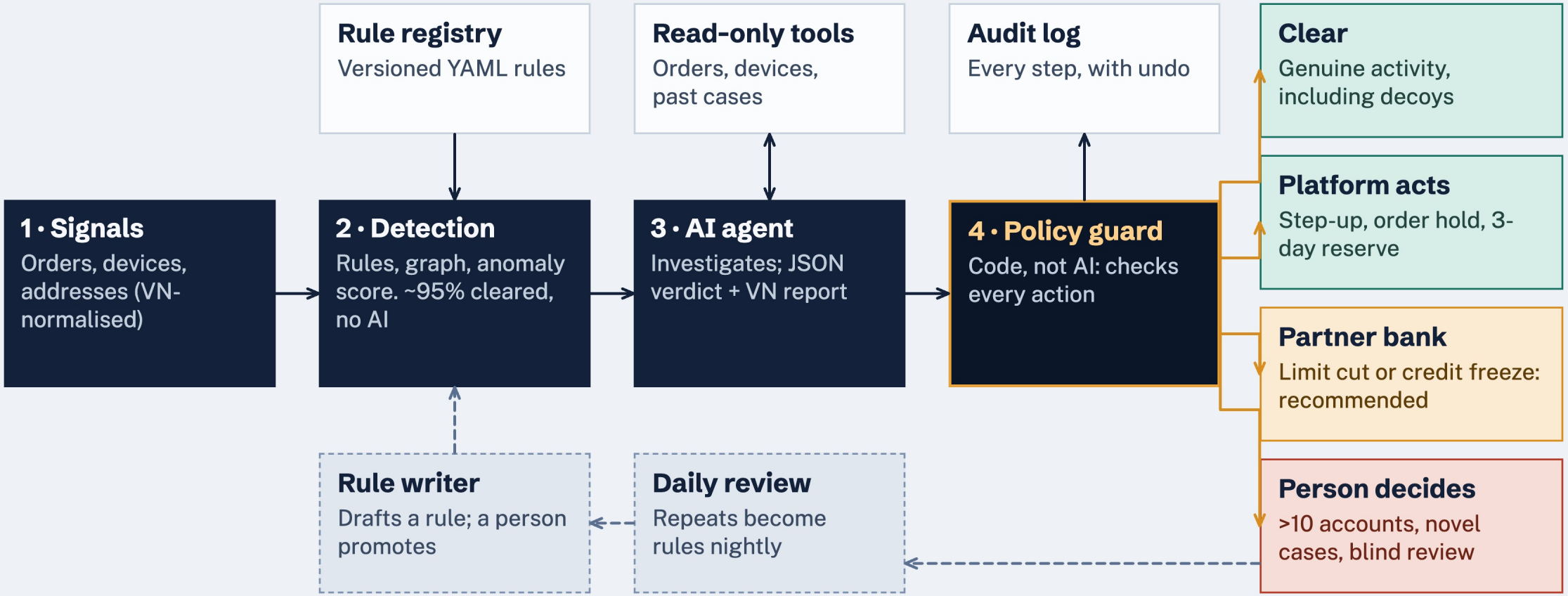
Each night it proposes rules for repeated decisions and flags similar cases that got different outcomes.



It learns only from **human decisions and confirmed outcomes**, never from its own verdicts.

AI flow design

Cheap signals first, the agent for judgment, a code guard before any action.



Solid: detect and investigate · Orange: guarded actions · Dashed: learning from human decisions

Inside the investigator agent

STEP 1

Scope the case

Forms a hypothesis: ring, cash-out or takeover.

STEP 2

Gather evidence

Read-only tools: devices, orders, repayments, past cases.

STEP 3

Reason and draft

Weighs fraud against innocent reasons, like a shared nhà trọ.

STEP 4

Self-check

Every cited record ID must exist, or it retries, then escalates.

STEP 5

Hand to the guard

Never acts itself. It recommends; the guard decides.

```
{
  "decision": "hold_and_escalate",
  "pattern": "cash_out_colluding_shop",
  "risk": 0.91, "confidence": 0.88, "novel": false,
  "evidence": ["dev_2211", "ord_88410", "shop_517"],
  "report_vi": "14 tài khoản, 3 thiết bị…"
}
```

A verdict, never an action

Structured outputs and tool calling via the OpenAI Responses API. Every claim cites a real record ID, so the code guard can check it before anything happens.

ADAPTIVE BY DESIGN

When it meets a new tactic, it adapts

1. **Agent flags novel: true**
2. **Analyst confirms**
3. **Rule writer drafts a rule**
4. **Backtest on history**
5. **A person promotes it**
6. **Next wave caught, no AI call**

Model routing

A fast model triages flagged cases; a reasoning model handles only rings and novel patterns. Most traffic never reaches the AI.

Knows when it doesn't know

If a case matches no known pattern, it says so. The guard then allows only mild, reversible actions and escalates.

Trusted because it knows what it may not do

Action	Who decides	Limit
Step-up verification	**Agent alone**	OTP for low risk, face match for high risk
Hold an order or reserve part of a payout	**Agent alone**	Expires within 3 working days, then released or escalated
Cut a limit or freeze credit	**Partner bank**	The bank is the lender; the agent recommends with evidence
Anything on more than 10 accounts, or a novel case	**Person**	Analyst decides before seeing the AI's call

Kill switch: recommend-only mode

Circuit breaker on action spikes

Neutral messages with a complaint link

Designed around Vietnam's AML, consumer protection and AI laws

What we'll show in 3 minutes

0:00 **A ring no single order reveals**
14 accounts sharing devices. Orders held automatically; the credit freeze goes to the bank.

0:45 **The decoy**
A family sharing one phone and a busy nhà trọ. The desk clears them; a blunt rule would not.

1:20 **A new cash-out tactic, live**
No rule fires. The graph catches it; a rule is drafted, backtested and promoted; the next wave is caught.

2:20 **Why you can trust it**
Audit trail, a consistency flag between two analysts, and the kill switch.

Targets on our synthetic test set

Catch rate on the new tactic, after adapting

0 → 90%+

Fraud caught before payout	**Measured live**
Genuine users blocked	**0**
Time per case vs manual	**Measured live**
Same case, same decision	**Measured live**
Cases resolved without AI	**~95%**

Targets, not results. All numbers are computed live on synthetic data.

The team

Technical PM · Team Lead

Thao

- 4+ years as a Technical Product Manager on complex platforms, building fintech solutions
- Founded an AML triage agent, Runner-Up at Agentic AI Build Week 2026
- Writes on fintech and compliance: build compliance in from day one

Role: leads the team; turns fraud rules into specs and tests; fraud scenarios, user flows, demo

Technical Lead · Full Stack

Thành

- 9+ years in backend and distributed systems; Tech Lead at SkyLab
- Built COSAP's layered fraud and anomaly detection: rules, statistics, LLM review
- Kafka billing pipeline at 100,000+ events per minute

Role: investigation backend, orchestration, evidence checks, action limits

AI Engineer

Hiếu

- 3+ years building production LLM systems and agent harnesses
- Shipped a company-wide harness: tool registry, sandboxed execution, memory, tracing
- Strength: tool-call reliability and OpenTelemetry evals

Role: agent runtime, rule writer, anomaly scoring, evals, review UI

We have shipped fintech together at SkyLab, from Lumicap to COSAP. **Codex is our fourth engineer.**